

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 08

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Activity 01

#include <stdio.h> - Imports standard input and output functions

#include <sys/ipc.h> - Imports definitions for inter process communication mechanisms

#include <sys/shm.h> - Imports functions for shared memory operations

#include <string.h> - Imports string related operations

int main() { - Entry point of the program

key_t key = 1234; - Creates unique key to identify the shared memory segment

int shmid; - Variable to store the shared memory ID

char *ptr; - Pointer which points the shared memory location

shmid = shmget(key, 4096, 0666 | IPC_CREAT); - Creates a shared memory which size is 4096 bits, permissions to read and write for all.

ptr = (char *) shmat(shmid, NULL, 0); - Attach shared memory to the process and returns pointer to the memory

sprintf(ptr, "Hello from shared memory!"); - write a sting in to shared memory

printf("Data written to shared memory\n"); - Print the confirmation message

shmdt(ptr); - Detach shared memory from the process

return 0;

}

Activity 02

#include <stdio.h> - Imports standard input and output functions

#include <sys/ipc.h> - Imports definitions for inter process communication mechanisms

#include <sys/shm.h> - Imports functions for shared memory operations

int main() { - Entry point of the program

key_t key = 1234; - Creates unique key to identify the shared memory segment

int shmid; - Variable to store the shared memory ID

char *ptr; - Pointer which points the shared memory location

shmid = shmget(key, 4096, 0666); - Access the existing shared memory located with key

ptr = (char *) shmat(shmid, NULL, 0); - Attach shared memory

printf("Read from shared memory: %s\n", ptr); - Read and print the message stored in shared memory

shmdt(ptr); - Detach shared memory

shmctl(shmid, IPC_RMID, NULL); -Detach the shared memory segment from the system

return 0;

}

Activity 03

```
GNU nano 7.2 Lab08.c
#include <stdio.h>
#include <sys/ipc.h>
#include <sys/shm.h>
#include <string.h>

int main() {
    key_t key = 1234;
    int shmid;
    char *ptr;

    shmid = shmget(key, 4096, 0666 | IPC_CREAT);
    ptr = (char *) shmat(shmid, NULL, 0);

    strcpy(ptr, "Operating System Lab");
    printf("Message written to shared memory\n");

    printf("Message read from the shared memory: %s\n", ptr);

    shmdt(ptr);

    shmctl(shmid, IPC_RMID, NULL);
    return 0;
}
```

```
vboxuser@Ubuntu: ~
vboxuser@Ubuntu:~$ nano Lab08.c
vboxuser@Ubuntu:~$ gcc -o Lab08 Lab08.c
vboxuser@Ubuntu:~$ ./Lab08
Message written to shared memory
Message read from the shared memory: Operating System Lab
vboxuser@Ubuntu:~$
```